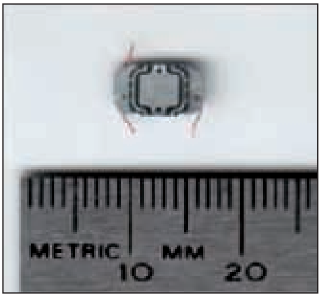
like a small grain of sand. Despite its size, the mu-chips are provided with 128 bits of ROM and this enables them to store 38-digit numbers. In fact, earlier this year, researchers at the Bristol University managed to successfully glue micro-transponders onto a colony of live ants. This helped them study their behaviour and consequently understand ant-hill dynamics. The only problem with having such small chips is that the ability to read is constrained by the inverse-square law. Thus to give it a large range, the signal strength will need to be made quite high.



PayPass RFID chip removed from a MasterCard

RFID tags are also becoming extremely cheap, and this further makes them a viable resource. In fact, earlier this year, Envego released a tag that costs only 5.9 cents a tag (Rs. 2.50 a tag).

The largest employer of RFID tags in the world is the USDOD (United States Department of Defence). They use RFID tags to tag every single container that goes out of the continental United States (CONUS).

There are however, quite a few controversies involving RFID. There are many organisations that have raised privacy issues.

> RFID systems have been accused of invading employee privacy

The chief concern for many is the fact that, at times, people may not be aware that they have been tagged. Also, a world-readable tag can make them prime prey to malicious intent. Another concern is that often the things people buy may be radio tagged. The size of the tag makes it impossible to detect, and thus these items may be used to carry out surveillance on the unsuspecting victim. It is also possible to program machines that read the identity from the tags, thus severely compromising security. Another, more technical, fear is that it is unknown what repeated exposure to radio waves can do to a person's physiology. There is no doubt that RFID has